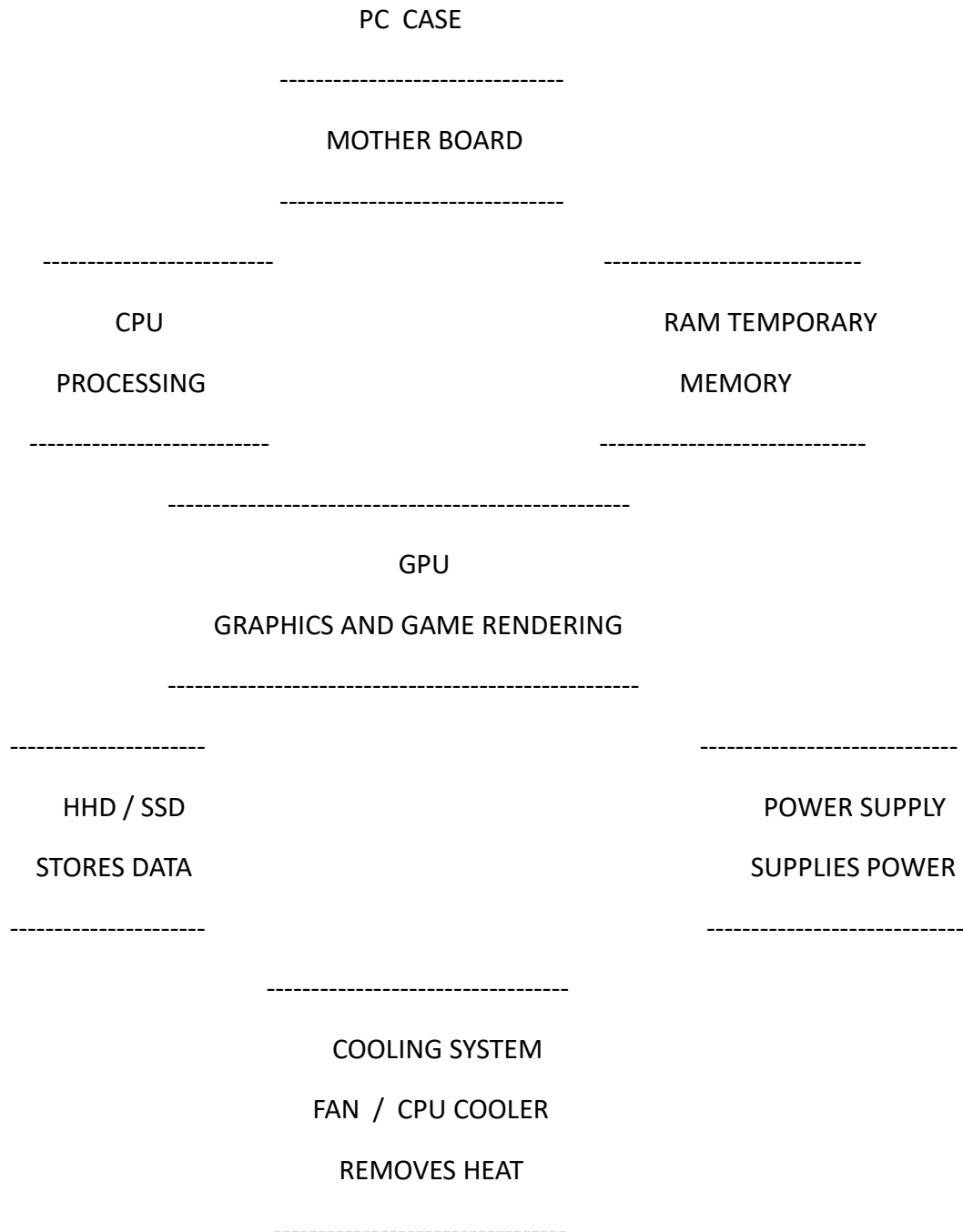


HARDWARE AND COMPONENT

1. Draw a labeled diagram of a desktop PC showing the motherboard, CPU, RAM, GPU, storage (HDD/SSD), power supply, and cooling system. For each component, write its main function next to it.



Component	Main Function
Motherboard	Connects all computer components and allows them to communicate with each other.
CPU (Central Processing Unit)	Processes instructions and performs calculations; it is the main processor of the computer.
RAM (Random Access Memory)	Temporarily stores data and programs that are currently being used, helping the PC work quickly.
GPU (Graphics Processing Unit)	Processes graphics, images, and video for the monitor and applications such as games.
HDD (Hard Disk Drive)	Provides long-term storage for files, programs, documents, and other data.
SSD (Solid-State Drive)	Provides fast long-term storage and helps the computer boot and load applications quickly.
Power Supply (PSU)	Converts electricity from the wall outlet into the required power for the PC components.
Cooling System	Removes heat from components such as the CPU and GPU to prevent overheating.

2. Create a table listing each major PC component (motherboard, CPU, RAM, GPU, storage, power supply, cooling system), its typical location inside the case, and one example of how it connects or interacts with another component.

Hint: For example, "CPU sits in the motherboard socket and communicates with RAM via the memory slots."

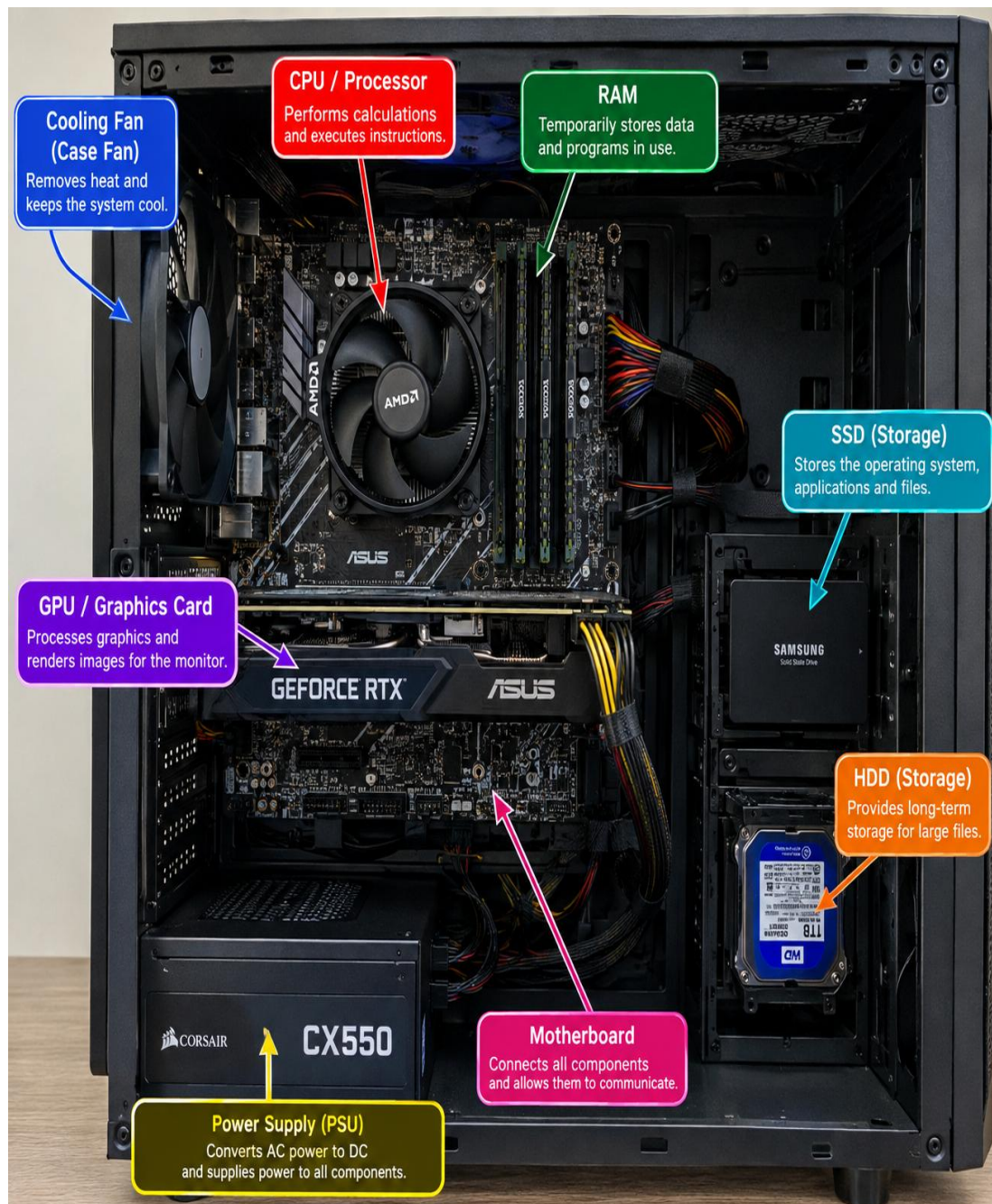
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PC Component	Typical Location Inside the Case	Example of Connection / Interaction
Motherboard	Mounted on the main side panel of the case	Connects the CPU, RAM, GPU, storage, and other components together.

PC Component	Typical Location Inside the Case	Example of Connection / Interaction
CPU	Installed in the CPU socket on the motherboard	Communicates with RAM through the motherboard's memory slots.
RAM	Installed in the DIMM slots next to the CPU	Exchanges data with the CPU for currently running programs.
GPU	Usually installed in a PCIe slot on the motherboard	Receives data from the CPU and sends processed graphics to the monitor.
Storage (HDD/SSD)	HDDs are usually in drive bays; SSDs may be mounted in 2.5-inch bays or M.2 slots	Connects to the motherboard through SATA or an M.2/PCIe interface.
Power Supply (PSU)	Usually at the bottom or sometimes the top/rear of the case	Supplies electrical power to the motherboard, GPU, storage, and other components.
Cooling System	CPU cooler sits directly above the CPU; case fans are mounted around the case	Removes heat from the CPU and circulates air through the case.

3 Take a clear photo or screenshot of the inside of any desktop PC (your own, a friend's, or an image from the web) and use any image editor to digitally label at least 6 major components as discussed in class.

-> I used a clear image of the inside of a desktop PC and digitally labeled the following major components:



1. **Motherboard** – Main circuit board that connects all components.
2. **CPU** – Processes instructions and performs calculations.
3. **RAM** – Temporarily stores data and programs currently in use.
4. **GPU (Graphics Card)** – Processes graphics and sends images to the monitor.
5. **Storage (HDD/SSD)** – Stores the operating system, applications, and files.

6. **Power Supply (PSU)** – Provides electrical power to all PC components.
7. **Cooling System** – Removes heat and prevents components from overheating.

Suggested labels for the screenshot:

Motherboard | CPU | RAM | GPU | HDD/SSD | Power Supply (PSU) | Cooling Fan

4. Write a short explanation (3-4 sentences) describing the data flow when you open a game on your PC, mentioning the roles of CPU, RAM, storage, GPU, and motherboard.

***Hint:** Start from storage loading data, then CPU processing, RAM usage, and GPU rendering.*

->When you open a game, the **storage (HDD/SSD)** loads the game files and sends them to **RAM** for quick access. The **CPU** processes the game instructions and sends the required data through the **motherboard** to other components. The **GPU** processes and renders the graphics, while the CPU and GPU work with RAM to keep the game running smoothly. The **motherboard** connects all these components and allows them to communicate with each other.